Unit-V

Correlation and Regression

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Probability & Statistics

Unit-V

Correlation and Regression

Curve Fitting
Least Squares Curve Fitting Procedure

Working procedure
To fit the straight line $y = a + bx$

(i) Form the normal equations

$$\sum y = na + b\sum x$$
$$\sum xy = a\sum x + b\sum x^2$$

(ii) Solve these normal equations as simultaneous equations for a, b.

(iii) Substitute the values of a, b in $y = a + bx$, which is the required line of best fit.

To fit the parabola $y = a + bx + cx^2$

(i) Form the normal equations

$$\sum y = na + b\sum x + c\sum x^2$$
$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$$
$$\sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4$$

(ii) Solve these normal equations as simultaneous equations for a, b, c.

(iii) Substitute the values of a, b, c in $y = a + bx + cx^2$, which is the required parabola of best fit.

Examples

Example 1: Fit a straight line $y = a + bx$ for the following data

x	0	5	10	15	20	25
y	12	15	17	22	24	30

Solution. Here $n = 6$.

Normal equations

$$\sum y = na + b\sum x$$
$$\sum xy = a\sum x + b\sum x^2$$

x	y	x^2	xy
0	12	0	0
5	15	25	75
10	17	100	170
15	22	225	330
20	24	400	480
25	30	625	750

$\sum x = 75$	$\sum y = 120$	$\sum x^2 = 1375$	$\sum xy = 1805$

Normal equations become

$$120 = 6a + 75b$$
$$1805 = 75a + 1375b$$

Solving these equations, we get $a = 11.2857, b = 0.6971$

Hence the line of best fit is $y = 11.2857 + 0.6971x$.

Example 2: A chemical company, wishing to study the effect of extraction time on the efficiency of an extraction operation, obtained the data shown in the following table

Extraction time minutes(x)	27	45	41	19	3	39	19	49	15	31
Efficiency (y)	57	64	80	46	62	72	52	77	57	68

Fit a straight line to the given data by the method of least squares.

Solution. Here $n = 10$.

Normal equations

$$\sum y = na + b\sum x$$
$$\sum xy = a\sum x + b\sum x^2$$

x	y	x^2	xy
27	57	729	1539
45	64	2025	2880
41	80	1681	3280
19	46	361	874
3	62	9	186
39	72	1521	2808
19	52	361	988
49	77	2401	3773
15	57	225	855
31	68	961	2108
$\sum x = 288$	$\sum y = 635$	$\sum x^2 = 10274$	$\sum xy = 19291$

Normal equations become

$$635 = 10a + 288b$$
$$19291 = 288a + 10274b$$

Solving these equations, we get $a = 48.9080, b = 0.5067$

Hence the line of best fit is $y = 48.9080 + 0.5067x$.

Example 3: Fit a second degree polynomial to the following data by the method of least squares

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

Solution. Here $n = 5$.

Let the equation of the second-degree polynomial be $y = a + bx + cx^2$ ... (1)

Normal equations are

$$\sum y = na + b\sum x + c\sum x^2$$

$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$$
$$\sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4$$

x	y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	1.8	1	1	1	1.8	1.8
2	1.3	4	8	16	2.6	5.2
3	2.5	9	27	81	7.5	22.5
4	6.3	16	64	256	25.2	100.8
$\sum x = 10$	$\sum y = 12.9$	$\sum x^2 = 30$	$\sum x^3 = 100$	$\sum x^4 = 354$	$\sum xy = 37.1$	$\sum x^2y = 130.3$

Now normal equations become

$$12.9 = 5a + 10b + 30c$$
$$37.1 = 10a + 30b + 100c$$
$$130.3 = 30a + 100b + 354c$$

Solving these equations, we get $a = 1.42, b = -1.07, c = 0.55$
Substituting a, b, c in (1) we get $y = 1.42 - 1.07x + 0.55x^2$.

Example 3: Fit a parabola to the following data

x	10	12	15	23	20
y	14	17	23	25	21

Solution. Here $n = 5$.
Let the equation of the second-degree polynomial be $y = a + bx + cx^2$ ... (1)
Normal equations are

$$\sum y = na + b\sum x + c\sum x^2$$
$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$$
$$\sum x^2y = a\sum x^2 + b\sum x^3 + c\sum x^4$$

x	y	x^2	x^3	x^4	xy	x^2y
10	14	100	1000	10000	140	1400
12	17	144	1728	20736	204	2448
15	23	225	3375	50625	345	5175
23	25	529	12167	279841	575	13225
20	21	400	8000	160000	420	8400
$\sum x = 80$	$\sum y = 100$	$\sum x^2 = 1398$	$\sum x^3 = 26270$	$\sum x^4 = 521202$	$\sum xy = 1684$	$\sum x^2y = 30648$

Now normal equations become

$$100 = 5a + 80b + 1398c$$

$$1684 = 80a + 1398b + 26270c$$
$$30648 = 1398a + 26270b + 521202c$$

Solving these equations, we get $a = -8.7279, b = 3.0099, c = -0.0695$
Substituting a, b, c in (1) we get $y = -8.7279 + 3.0099x - 0.0695x^2$.

Fitting an exponential curve

(i) Consider a curve $y = ae^{bx}$

Taking logarithms on both sides, $log_{10}y = log_{10}[ae^{bx}]$

$$\Rightarrow log_{10}y = log_{10}a + log_{10}e^{bx}$$
$$\Rightarrow log_{10}y = log_{10}a + bxlog_{10}e$$
$$i.e., Y = A + Bx, \text{ where } Y = log_{10}y, A = log_{10}a, B = blog_{10}e$$

Here the normal equations are

$$\sum Y = nA + B\sum x$$
$$\sum xY = A\sum x + B\sum x^2$$

Solving these equations, we get A and B.

Since $A = log_{10}a \Rightarrow a = 10^A$ and $B = blog_{10}e \Rightarrow b = \frac{B}{log_{10}e}$

Substitute a and b in $y = ae^{bx}$.

Examples

Example 1: Determine the constants a and b by the method of least squares such that $y = ae^{bx}$.

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

Solution. Given $y = ae^{bx}$

Taking logarithms on both sides, $log_{10}y = log_{10}[ae^{bx}]$

$$\Rightarrow log_{10}y = log_{10}a + log_{10}e^{bx}$$
$$\Rightarrow log_{10}y = log_{10}a + bxlog_{10}e$$
$$i.e., Y = A + Bx, \text{ where } Y = log_{10}y, A = log_{10}a, B = blog_{10}e$$

Here the normal equations are

$$\sum Y = nA + B\sum x$$
$$\sum xY = A\sum x + B\sum x^2$$

Here $n = 5$

x	y	$Y = log_{10}y$	xY	x^2
2	4.077	0.6103	1.2206	4
4	11.084	1.0447	4.1788	16
6	30.128	1.4790	8.8740	36
8	81.897	1.9133	15.3064	64
10	222.62	2.3476	23.4760	100
$\sum x = 30$		$\sum Y = 7.3948$	$\sum xY = 53.0558$	$\sum x^2 = 220$

Normal equations:

$$7.3948 = 5A + 30B$$
$$53.0558 = 30A + 220B$$

Solving these equations, we get $A = 0.1759, B = 0.2172$

$A = log_{10}a \Rightarrow a = 10^A \Rightarrow a = 10^{0.1759} \Rightarrow a = 1.4993$

and $B = blog_{10}e \Rightarrow b = \frac{B}{log_{10}e} \Rightarrow b = \frac{0.2172}{0.4343} \Rightarrow b = 0.5001$

Hence, $y = 1.4993e^{0.5001x}$ is the required curve.

Fitting an exponential curve

(ii) Consider a curve $y = ab^x$

Taking logarithms on both sides, $log_{10}y = log_{10}[ab^x]$

$$\Rightarrow log_{10}y = log_{10}a + log_{10}b^x$$
$$\Rightarrow log_{10}y = log_{10}a + xlog_{10}b$$
$$i.e., Y = A + Bx, \text{ where } Y = log_{10}y, A = log_{10}a, B = log_{10}b$$

Here the normal equations are

$$\sum Y = nA + B\sum x$$
$$\sum xY = A\sum x + B\sum x^2$$

Solving these equations, we get A and B.

Since $A = log_{10}a \Rightarrow a = 10^A$ and $B = log_{10}b \Rightarrow b = 10^B$

Substitute a and b in $y = ab^x$.

Example 1: Obtain a relation of the form $y = ab^x$ for the following data by the method of least squares

x	2	3	4	5	6
y	8.3	15.4	33.1	65.2	127.4

Solution. Given $y = ab^x$

Taking logarithms on both sides, $log_{10}y = log_{10}[ab^x]$

$$\Rightarrow log_{10}y = log_{10}a + log_{10}b^x$$
$$\Rightarrow log_{10}y = log_{10}a + xlog_{10}b$$
$$i.e., Y = A + Bx, \text{ where } Y = log_{10}y, A = log_{10}a, B = log_{10}b$$

Here the normal equations are

$$\sum Y = nA + B\sum x$$
$$\sum xY = A\sum x + B\sum x^2$$

Here $n = 5$

$\boldsymbol{x}$	$\boldsymbol{y}$	$\boldsymbol{Y = log_{10}y}$	$\boldsymbol{xY}$	$\boldsymbol{x^2}$
2	8.3	0.9191	1.8382	4
3	15.4	1.1875	3.5625	9
4	33.1	1.5198	6.0792	16
5	65.2	1.8142	9.0710	25
6	127.4	2.1052	12.6312	36
$\sum x = 20$		$\sum Y = 7.5458$	$\sum xY = 33.1821$	$\sum x^2 = 90$

Normal equations:

$$7.5458 = 5A + 20B$$

$$33.1821 = 20A + 90B$$

Solving these equations, we get $A = 0.3096, B = 0.2999$

$A = log_{10}a \Rightarrow a = 10^A \Rightarrow a = 10^{0.3096} \Rightarrow a = 2.0399$

and $B = log_{10}b \Rightarrow b = 10^B \Rightarrow b = 10^{0.2999} \Rightarrow b = 1.9948$

Hence, $y = 2.0399(1.9948)^x$ is the required curve.

Fitting a power curve

(iii) Consider a curve $y = ax^b$

Taking logarithms on both sides, $log_{10}y = log_{10}[ax^b]$

$$\Rightarrow log_{10}y = log_{10}a + log_{10}x^b$$
$$\Rightarrow log_{10}y = log_{10}a + blog_{10}x$$
$$i.e., Y = A + bX, \text{ where } Y = log_{10}y, A = log_{10}a, X = log_{10}x$$

Here the normal equations are

$$\sum Y = nA + b\sum X$$
$$\sum XY = A\sum x + b\sum X^2$$

Solving these equations, we get A and b.

Since $A = log_{10}a \Rightarrow a = 10^A$

Substitute a and b in $y = ax^b$.

Example 1: Fit a power curve $y = ax^b$ for the following data

x	6	2	10	5	8
y	9	11	12	8	7

Solution. Given $y = ab^x$

Taking logarithms on both sides, $log_{10}y = log_{10}[ax^b]$

$$\Rightarrow log_{10}y = log_{10}a + log_{10}x^b$$
$$\Rightarrow log_{10}y = log_{10}a + blog_{10}x$$
$$i.e., Y = A + bX, \text{ where } Y = log_{10}y, A = log_{10}a, X = log_{10}x$$

Here the normal equations are

$$\sum Y = nA + b\sum X$$
$$\sum XY = A\sum x + b\sum X^2$$

x	y	$X = log_{10}x$	$Y = log_{10}y$	XY	X^2
6	9	0.7782	0.9542	0.7425	0.6055
2	11	0.3010	1.0414	0.3135	0.0906
10	12	1.0000	1.0792	1.0792	1.0000
5	8	0.6990	0.9031	0.6312	0.4886
8	7	0.9031	0.8451	0.762=32	0.8156
		$\sum X = 3.6813$	$\sum Y = 4.8230$	$\sum XY = 3.5296$	$\sum X^2 = 3.0003$

Normal equations: Here $n = 5$

$$4.8230 = 5A + 3.6813b$$
$$3.5296 = 3.6813A + 3.0003b$$

Solving these equations, we get $A = 1.0189, b = -0.0738$

$A = log_{10}a \Rightarrow a = 10^A \Rightarrow a = 10^{1.0189} \Rightarrow a = 10.4448$

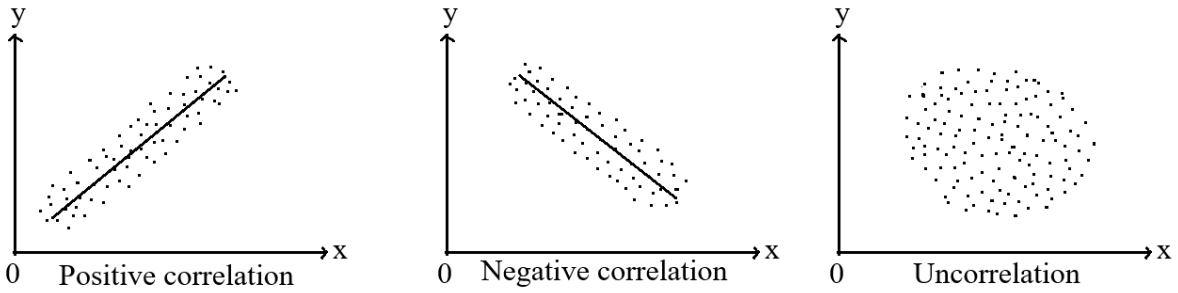
Hence, $y = 10.4448x^{-0.0738}$ is the required curve.

Correlation

Definition. Correlation is a statistical analysis which measures and analyses the degree or extent to which two variables fluctuate with reference to each other.

If there exists some relationship between two variables then the statistical analysis of such data is called bivariate analysis. For example, there is some relation between the price and the sale, price and the production of a commodity.

If an increase (or decrease) in the values of one variable corresponds to an increase (or decrease) in the other, the correlation is said to be **positive**. If the increase (or decrease) in one corresponds to the decrease (or increase) in the other, the correlation is said to be **negative**. If there is no relationship indicated between the variables, they are said to be **independent** or **uncorrelated**.



Coefficient of correlation (Karl Pearson's)

The numerical measure of correlation is called the coefficient of correlation and is defined by the relation

$$r = \frac{\sum XY}{n\sigma_x\sigma_y}$$

where $X = x - \overline{x}, Y = y - \overline{y}$,
σ_x = standard deviation of x −series,
σ_y = standard deviation of y −series, and
n = number of values of the two variables.

Methods of calculation

1. $r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[\{n\sum x^2 - (\sum x)^2\}\{n\sum y^2 - (\sum y)^2\}]}}$ (General)

2. $r = \frac{\sum XY}{\sqrt{(\sum X^2)(\sum Y^2)}}, where\ X = x - \overline{x}, Y = y - \overline{y}$
   $(Here\ \sum X = 0, \sum Y = 0)$

3. Step-deviation method $r = \frac{n\sum d_x d_y - \sum d_x \sum d_y}{\sqrt{\left[\{n\sum d_x^2 - (\sum d_x)^2\}\left\{n\sum d_y^2 - (\sum d_y)^2\right\}\right]}}$ $(\overline{x}, \overline{y}\ are\ not\ integers)$
   where $d_x = \frac{x-a}{h}$, a = assumed mean of x or $d_x = x - a$,
   $d_y = \frac{y-b}{k}$, b = assumed mean of y or $d_y = y - b$.

Note: The coefficient of correlation lies between -1 and 1 i.e., $-1 \leq r \leq 1$.

Examples

Example 1: Calculate the coefficient of correlation from the following data

x	12	9	8	10	11	13	7
y	14	8	6	9	11	12	3

Solution.

Here we use

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[\{n\sum x^2 - (\sum x)^2\}\{n\sum y^2 - (\sum y)^2\}]}}$$

(since x, y are small values)

Here $n = 7$

$\boldsymbol{x}$	$\boldsymbol{y}$	$\boldsymbol{x^2}$	$\boldsymbol{y^2}$	$\boldsymbol{xy}$
12	14	144	196	168
9	8	81	64	72
8	6	64	36	48
10	9	100	81	91
11	11	121	121	121
13	12	169	144	156
7	3	49	9	21
$\sum x = 70$	$\sum y = 63$	$\sum x^2 = 728$	$\sum x^2 = 651$	$\sum xy = 676$

$$r = \frac{7(676) - (70)(63)}{\sqrt{[\{7(728) - (70)^2\}\{7(651) - (63)^2\}]}} = 0.95$$

Example 2: Find Karl Pearson's coefficient of correlation from the following data

Wages	100	101	102	102	100	99	97	98	96	95
Cost of living	98	99	99	97	95	92	95	94	90	91

Solution. Here we use

$$r = \frac{\sum XY}{\sqrt{(\sum X^2)(\sum Y^2)}}, where\ X = x - \overline{x}, Y = y - \overline{y}$$

$$\overline{x} = \frac{100 + 101 + 102 + 102 + 100 + 99 + 97 + 98 + 96 + 95}{10} = \frac{990}{10} = 99$$

$$\overline{y} = \frac{98 + 99 + 99 + 97 + 95 + 92 + 95 + 94 + 90 + 91}{10} = \frac{950}{10} = 95$$

Wages $(\boldsymbol{x})$	$\boldsymbol{X = x - \overline{x}}$	$\boldsymbol{X^2}$	**Cost of living** $(\boldsymbol{y})$	$\boldsymbol{Y = y - \overline{y}}$	$\boldsymbol{Y^2}$	$\boldsymbol{XY}$
100	1	1	98	3	9	3
101	2	4	99	4	16	8
102	3	9	99	4	16	12
102	3	9	97	2	4	6
100	1	1	95	0	0	0
99	0	0	92	-3	9	0
97	-2	4	95	0	0	0
98	-1	1	94	-1	1	1
96	-3	9	90	-5	25	15
95	-4	16	91	-4	16	16
$\sum x = 990$	$\sum X = 0$	$\sum X^2 = 54$	$\sum y = 950$	$\sum Y = 0$	$\sum Y^2 = 96$	$\sum XY = 61$

$$r = \frac{\sum XY}{\sqrt{(\sum X^2)(\sum Y^2)}} = \frac{61}{\sqrt{54 \times 96}} = 0.85$$

Example 3: Psychological test of intelligence and of engineering ability were applied to 10 students. Here is a record of ungrouped data showing intelligence ratio (I.R.) and engineering ratio (E.R.). Calculate the coefficient of correlation

Student	A	B	C	D	E	F	G	H	I	J
I.R.	105	104	102	101	100	99	98	96	93	92
E.R.	101	103	100	98	95	96	104	92	97	94

Solution.

Here we use

$$r = \frac{\sum XY}{\sqrt{(\sum X^2)(\sum Y^2)}}, where\ X = x - \overline{x}, Y = y - \overline{y}$$

I.R. (x): Mean of x

$$\overline{x} = \frac{105 + 104 + 102 + 101 + 100 + 99 + 98 + 96 + 93 + 92}{10} = \frac{990}{10} = 99$$

E.R. (y): Mean of y

$$\overline{y} = \frac{101 + 103 + 100 + 98 + 95 + 96 + 104 + 92 + 97 + 94}{10} = \frac{980}{10} = 98$$

Student	**I.R.** (x)	$X = x - \overline{x}$	X^2	**E.R.** (y)	$Y = y - \overline{y}$	Y^2	XY
A	105	6	36	101	3	9	18
B	104	5	25	103	5	25	25
C	102	3	9	100	2	4	6
D	101	2	4	98	0	0	0
E	100	1	1	95	-3	9	-3
F	99	0	0	96	-2	4	0
G	98	-1	1	104	6	36	-6
H	96	-3	9	92	-6	36	18
I	93	-6	36	97	-1	1	6
J	92	-7	49	94	-4	16	28
	$\sum x = 990$	$\sum X = 0$	$\sum X^2 = 170$	$\sum y = 950$	$\sum Y = 0$	$\sum Y^2 = 140$	$\sum XY = 92$

$$r = \frac{\sum XY}{\sqrt{(\sum X^2)(\sum Y^2)}} = \frac{92}{\sqrt{170 \times 140}} = 0.59$$

Example 4: Find out the correlation coefficient in the following case:

Height of father (in inches)	65	66	67	67	68	69	71	73
Height of son (in inches)	67	68	64	68	72	70	69	70

Solution.

Height of father (x): Mean: $\overline{x} = \frac{65+66+67+67+68+69+71+73}{8} = 68.25$ (not an integer)

Height of son (y): Mean: $\overline{y} = \frac{67+68+64+68+72+70+69+70}{8} = 68.5$ (not an integer)
Here $n = 8$,
assumed mean of $x, a = 68$ (one of x values) and
assumed mean of $y, b = 68$ (one of y values)

Height of father (x)	$\boldsymbol{d_x = x - a}$	$\boldsymbol{d_x^2}$	**Height of son (y)**	$\boldsymbol{d_y = y - b}$	$\boldsymbol{d_y^2}$	$\boldsymbol{d_x d_y}$
65	-3	9	67	-1	1	3
66	-2	4	68	0	0	0
67	-1	1	64	-4	16	4
67	-1	1	68	0	0	0
68	0	0	72	4	16	0
69	1	1	70	2	4	2
72	3	9	69	1	1	3
73	5	25	70	2	4	10
	$\sum d_x = 2$	$\sum d_x^2 = 50$		$\sum d_y = 4$	$\sum d_y^2 = 42$	$\sum d_x d_y = 22$

Here we use

$$r = \frac{n\sum d_x d_y - \sum d_x \sum d_y}{\sqrt{\left[\{n\sum d_x^2 - (\sum d_x)^2\}\left\{n\sum d_y^2 - \left(\sum d_y\right)^2\right\}\right]}}$$

$$= \frac{8(22) - (2)(4)}{\sqrt{[\{8(50) - 2^2\}\{8(42) - 4^2\}]}} = 0.472$$

Example 5: Find a suitable coefficient of correlation for the following data:

Fertiliser used, x (tonnes)	15	18	20	24	30	35	40	50
Productivity, y (tonnes)	85	93	95	105	120	130	150	160

Solution.
Mean of x: $\overline{x} = \frac{15+18+20+24+30+35+40+50}{8} = 29$
Mean of y: $\overline{y} = \frac{85+93+95+105+120+130+150+160}{8} = 117.25$ (not an integer)
Here $n = 8$,
assumed mean of $x, a = 29$ and
assumed mean of $y, b = 120$ (one of y values)

Fertiliser used (x)	$\boldsymbol{d_x = x - a}$	$\boldsymbol{d_x^2}$	**Productivity (y)**	$\boldsymbol{d_y = y - b}$	$\boldsymbol{d_y^2}$	$\boldsymbol{d_x d_y}$
15	-14	196	85	-35	1225	490
18	-11	121	93	-27	729	297
20	-9	81	95	-25	625	225
24	-5	25	105	-15	225	75
30	1	1	120	0	0	0

35	6	36	130	10	100	60
40	11	121	150	30	900	330
50	21	441	160	40	1600	840
	$\sum d_x = 0$	$\sum d_x^2 = 50$		$\sum d_y = -22$	$\sum d_y^2 = 5404$	$\sum d_x d_y = 2317$

Here we use

$$r = \frac{n\sum d_x d_y - \sum d_x \sum d_y}{\sqrt{\left[\{n\sum d_x^2 - (\sum d_x)^2\}\left\{n\sum d_y^2 - \left(\sum d_y\right)^2\right\}\right]}}$$

$$= \frac{8(2317) - (0)(-22)}{\sqrt{[\{8(1022) - 0^2\}\{8(5404) - (-22)^2\}]}} = 0.992$$

Rank Correlation

A group of n individuals may be arranged in order of merit with respect to some characteristic. The same group would give different order for different characteristics. Considering the orders corresponding to two characteristics A and B, the correlation between these n pairs of ranks is called the **rank correlation** in the characteristics A and B for that group of individuals. This method is based on rank and is useful in dealing with qualitative characteristics such as morality, character, intelligence and beauty.

The Spearman's rank correlation is given by

$$\rho = 1 - \frac{6\sum d^2}{n^3 - n}$$

where $d =$ rank of x - rank of y = difference of two ranks.

Properties of rank correlation coefficient

1. The value of ρ lies between -1 and 1.
2. If $\rho = 1$, there is complete agreement in the order if the ranks and the direction of the rank is same.
3. If $\rho = -1$, there is complete disagreement in the order of the ranks and they are in opposite directions.

Examples

Example 1: Following are the ranks obtained by 10 students in two subjects, statistics and mathematics. To what extent the knowledge of the students in two subjects is related? (Find the rank correlation coefficient ρ).

Statistics	1	2	3	4	5	6	7	8	9	10
Mathematics	2	4	1	5	3	9	7	10	6	8

Solution.

Rank in Statistics (x)	**Rank in Mathematics (y)**	$\boldsymbol{d = x - y}$	$\boldsymbol{d^2}$
1	2	-1	1
2	4	-2	4
3	1	2	4
4	5	-1	1
5	3	2	4
6	9	-3	9
7	7	0	0

8	10	-2	4
9	6	3	9
10	8	2	4
			$\sum d^2 = 40$

Here $n = 10$

Rank correlation coefficient is given by

$$\rho = 1 - \frac{6\sum d^2}{n^3 - n}$$
$$= 1 - \frac{6(40)}{10^3 - 10}$$
$$= 0.76$$

Example 2: Three judges A, B, C give the following ranks. Find which pair of judges has common approach

A	1	6	5	10	3	2	4	9	7	8
B	3	5	8	4	7	10	2	1	6	9
C	6	4	9	8	1	2	3	10	5	7

Solution.

$\boldsymbol{A}$ $(\boldsymbol{x})$	$\boldsymbol{B}$ $(\boldsymbol{y})$	$\boldsymbol{C}$ $(\boldsymbol{z})$	$\boldsymbol{d_1}$ $\boldsymbol{= x - y}$	$\boldsymbol{d_2}$ $\boldsymbol{= y - z}$	$\boldsymbol{d_3}$ $\boldsymbol{= z - x}$	$\boldsymbol{d_1^2}$	$\boldsymbol{d_2^2}$	$\boldsymbol{d_3^2}$
1	3	6	-2	-2	5	4	9	25
6	5	4	1	1	-2	1	1	4
5	8	9	-3	-1	4	9	1	16
10	4	8	6	-4	-2	36	16	4
3	7	1	-4	6	-2	16	63	4
2	10	2	-8	8	0	64	64	0
4	2	3	2	-1	-1	4	1	1
9	1	10	8	-9	1	64	81	1
7	6	5	1	1	-2	1	1	4
8	9	7	-1	2	-1	1	4	1
						$\sum d_1^2 = 200$	$\sum d_2^2 = 214$	$\sum d_3^2 = 60$

$$\rho(x, y) = 1 - \frac{6\sum d_1^2}{n^3 - n} = 1 - \frac{6(200)}{10^3 - 10} = -0.2$$

$$\rho(y, z) = 1 - \frac{6\sum d_2^2}{n^3 - n} = 1 - \frac{6(214)}{10^3 - 10} = -0.3$$

$$\rho(z, x) = 1 - \frac{6\sum d_3^2}{n^3 - n} = 1 - \frac{6(60)}{10^3 - 10} = 0.6$$

Since $\rho(z, x)$ is maximum, the pair of judges A, C have the nearest common approach.

Equal or repeated ranks

If there is more than one item with the same value in the series then common ranks are given to repeated items. The common rank is the average of the ranks which these items would have assumed, if they are different from each other then the next item will get the rank next to ranks already assumed.

For example, if two individuals are placed in the 7th place, each of them is given the rank 7.5 $\left(=\frac{7+8}{2}\right)$ and the next will be 9. Similarly, if 3 are ranked equal at the 7th place then they are given the rank 8 $\left(=\frac{7+8+9}{3}\right)$ which is the common rank assigned to each and the next rank will be 10.

Here, the Spearman's rank correlation coefficient is

$$\rho = 1 - \frac{6\left[\sum d^2 + \frac{m^3-m}{12} + \frac{m^3-m}{12} + \cdots\right]}{n^3-n}$$

where m is the number of items whose ranks are common.

Examples

Example 1: From the following data calculate the rank correlation coefficient after making adjustment for tied ranks

x	48	33	40	9	16	16	65	24	16	57
y	13	13	24	6	15	4	20	9	6	19

Solution.

First, we have to assign ranks to the variables (Smaller number is given rank 1 or bigger number is given rank 1). Here smaller number is given rank 1.

x	y	Rank of x	Rank of y	$d = rank\ of\ x - rank\ of\ y$	d^2
48	13	8	5.5	2.5	6.25
33	13	6	5.5	0.5	0.25
40	24	7	10	-3	9
9	6	1	2.5	-1.5	2.25
16	15	3	7	4	16
16	4	3	1	2	4
65	20	10	9	1	1
24	9	5	4	1	1
16	6	3	2.5	5	25
57	19	9	8	1	1
					$\sum d^2 = 41$

Here $n = 10$

x: 16 is repeated 3 times, so $m = 3$ and rank assigned is $\frac{2+3+4}{3} = 3$

y: 6 is repeated 2 times, so $m = 2$ and rank assigned is $\frac{2+3}{2} = 2.5$

y: 13 is repeated 2 times, so $m = 2$ and rank assigned is $\frac{5+6}{2} = 5.5$

Spearman's rank correlation coefficient is

$$\rho = 1 - \frac{6\left[\sum d^2 + \frac{m^3-m}{12} + \frac{m^3-m}{12} + \frac{m^3-m}{12}\right]}{n^3-n}$$

$$= 1 - \frac{6\left[41 + \frac{3^3-3}{12} + \frac{2^3-2}{12} + \frac{2^3-2}{12}\right]}{10^3-10}$$

$$= 0.733$$

Example 2: A sample of 12 fathers and their elder sons gave the following data about their elder sons. Calculate the coefficient of rank correlation

Fathers	65	63	67	64	68	62	70	66	68	67	69	71
Sons	68	66	68	65	69	66	68	65	71	67	68	70

Solution.

First, we have to assign ranks to the variables (Smaller number is given rank 1 or bigger number is given rank 1). Here bigger number is given rank 1.

Fathers x	**Sons** y	**Rank of** x	**Rank of** y	$d = rank\ of\ x - rank\ of y$	d^2
65	68	9	5.5	3.5	12.25
63	66	11	9.5	1.5	2.25
67	68	6.5	5.5	1.0	1
64	65	10	11.5	-1.5	2.25
68	69	4.5	3	1.5	2.25
62	66	12	9.5	2.5	6.25
70	68	2	5.5	-3.5	12.25
66	65	8	11.5	3.5	12.25
68	71	4.5	1	-3.5	12.25
67	67	6.5	8	-1.5	2.25
69	68	3	5.5	-2.5	6.25
71	70	1	2	-1	1
					$\sum d^2 = 72.5$

Here $n = 12$

x: 68 is repeated 2 times, so $m = 2$ and rank assigned is $\frac{4+5}{2} = 4.5$

x: 67 is repeated 2 times, so $m = 2$ and rank assigned is $\frac{6+7}{2} = 6.5$

y: 68 is repeated 4 times, so $m = 4$ and rank assigned is $\frac{4+5+6+7}{4} = 5.5$

y: 66 is repeated 2 times, so $m = 2$ and rank assigned is $\frac{9+10}{2} = 9.5$

y: 65 is repeated 2 times, so $m = 2$ and rank assigned is $\frac{11+12}{2} = 11.5$

Spearman's rank correlation coefficient is

$$\rho = 1 - \frac{6\left[\sum d^2 + \frac{m^3-m}{12} + \frac{m^3-m}{12} + \frac{m^3-m}{12} + \frac{m^3-m}{12} + \frac{m^3-m}{12}\right]}{n^3 - n}$$

$$= 1 - \frac{6\left[72.5 + \frac{2^3-3}{12} + \frac{2^3-2}{12} + \frac{4^3-4}{12} + \frac{2^3-2}{12} + \frac{2^3-2}{12}\right]}{12^3 - 12}$$

$$= 0.722$$

Regression

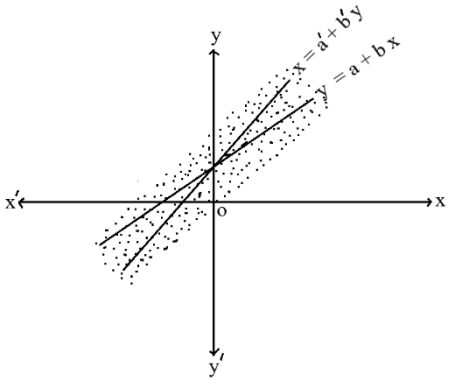
In regression, we can estimate the value of one variable with the value of the other variable which is known. The statistical method which helps us to estimate the unknown value of one variable from the known value of the related variable is called regression.

Distinction between Correlation and Regression

Sl No	Correlation	Regression
1	It measures the degree and direction of relationship between the variables.	It measures the nature and extent of average relationship between two or more variables in terms of the original units of the data
2	It is a relative measure showing association between the variables.	It is an absolute measure of relationship.
3	Correlation Coefficient is independent of change of both origin and scale.	Regression Coefficient is independent of change of origin but not scale.
4	Correlation Coefficient is independent of units of measurement.	Regression Coefficient is not independent of units of measurement.
5	Expression of the relationship between the variables ranges from –1 to +1.	Expression of the relationship between the variables may be in any of the forms like: Y = a + bX Y = a + bX + cX²
6	It is not a forecasting device.	It is a forecasting device which can be used to predict the value of dependent variable from the given value of independent variable.
7	There may be zero correlation such as weight of wife and income of husband.	There is nothing like zero regression.

Lines of Regression

It frequently happens that the dots of a scatter diagram generally, tend to cluster along a well-defined direction which suggests a linear relationship between the variables x and y. Such a line of best fit for the given distribution of dots is called the line of regression. In fact, there are two such lines, one giving the best possible mean values of y for each specified value of x and the other giving the best possible mean values of x for given values of y. The former is known as the line of regression of y on x and the latter as the line of regression of x on y.



The line of regression of y on x is $y - \overline{y} = r\frac{\sigma_y}{\sigma_x}(x - \overline{x})$.

or $y - \overline{y} = b_{yx}(x - \overline{x})$.

Its slope i.e., $b_{yx} = r\frac{\sigma_y}{\sigma_x}$ is called the regression coefficient of y on x.

The line of regression of x on y is $x - \overline{x} = r\frac{\sigma_x}{\sigma_y}(y - \overline{y})$

or $x - \overline{x} = b_{xy}(y - \overline{y})$

Its slope i.e., $b_{xy} = r\frac{\sigma_x}{\sigma_y}$ is called the regression coefficient of x on y.

Note: The correlation coefficient r is the geometric mean between the two regression coefficients i.e., $b_{yx} \times b_{xy} = r\frac{\sigma_y}{\sigma_x} \times r\frac{\sigma_x}{\sigma_y} = r^2$ or $r = \sqrt{r\frac{\sigma_y}{\sigma_x} \times r\frac{\sigma_x}{\sigma_y}}$.

Properties of Regression Coefficients:

1. The coefficient of correlation is the geometric mean of the two regression coefficients. Symbolically r = $\sqrt{bxy * byx}$
2. If one of the regression coefficients is greater than unity, the other must be less than unity, since the value of the coefficient of correlation cannot exceed unity. For example if bxy = 1.2 and byx = 1.4 "r" would be = $\sqrt{1.2 * 1.4}$ = 1.29, which is not possible.
3. Both the regression coefficient will have the same sign. i.e. they will be either positive or negative. In other words, it is not possible that one of the regression coefficients is having minus sign and the other plus sign.
4. The coefficient of correlation will have the same sign as that of regression coefficient, i.e. if regression coefficient have a negative sign, "r" will also have negative sign and if the regression coefficient have a positive sign, "r" would also be positive. For example, if b_{xy} = -0.2 and b_{yx} = -0.8 then r = - $\sqrt{0.2 * 0.8}$ = **– 0.4**
5. The average value of the two regression coefficient would be greater than the value of coefficient of correlation. In symbol (bxy + byx) / 2 > r. For example, if bxy = 0.8 and byx = 0.4 then average of the two values = (0.8 + 0.4) / 2 = 0.6 and the value of r = r = $\sqrt{0.8 * 0.4}$ = 0.566 which less than 0.6
6. Regression coefficients are independent of change of origin but not scale.

Examples

Example 1: Find the mean values of the variables x and y and correlation coefficient from the following regression equations $2y - x - 50 = 0, 3y - 2x - 10 = 0$.

Solution.

Since the means of $x's$ and the means of $y's$ lie on the two regression lines, we have

$$2\overline{y} - \overline{x} - 50 = 0 \ldots (1), 3\overline{y} - 2\overline{x} - 10 = 0 \ldots (2)$$

Solving these equations, we get $\overline{x} = 130, \overline{y} = 90$ (mean values)

Rewriting (1) and (2), we get

$\overline{y} = \frac{1}{2}\overline{x} + 50$ and $\overline{x} = \frac{3}{2}\overline{y} - 5$

Or $y = \frac{1}{2}x + 50$ and $x = \frac{3}{2}y - 5$

Therefore, regression coefficient of y on x is given by $r\frac{\sigma_y}{\sigma_x} = \frac{1}{2}$ and

regression coefficient of x on y is given by $r\frac{\sigma_x}{\sigma_y} = \frac{3}{2}$

Regression coefficient, $r = \sqrt{r\frac{\sigma_y}{\sigma_x} \times r\frac{\sigma_x}{\sigma_y}} = \sqrt{\frac{1}{2} \times \frac{3}{2}} = 0.866$

Example 2: The two regression equations of the variables x and y are $x = 19.13 - 0.87y, and\ \ y = 11.64 - 0.50x$. Find (i) mean of $x's$, (ii) mean of $y's$ and (iii) the correlation coefficient between x and y.

Solution.

Since the means of $x's$ and the means of $y's$ lie on the two regression lines, we have

$$\overline{x} = 19.13 - 0.87\overline{y} \ldots (1), \overline{y} = 11.64 - 0.50\overline{x} \ldots (2)$$

Solving these equations, we get $\overline{x} = 15.79, \overline{y} = 3.74$ (mean values)

From $\overline{y} = 11.64 - 0.50\overline{x}$, regression coefficient of y on x is given by $r\frac{\sigma_y}{\sigma_x} = -0.50$ and

From $\overline{x} = 19.13 - 0.87\overline{y}$, regression coefficient of x on y is given by $r\frac{\sigma_x}{\sigma_y} = -0.87$

Regression coefficient, $r = \sqrt{r\frac{\sigma_y}{\sigma_x} \times r\frac{\sigma_x}{\sigma_y}} = \sqrt{(-0.50)(-0.87)} = \sqrt{0.43} = -0.66$

[negative sign is taken since both the regression coefficients are negative]

Example 3: Calculate the regression equation of y on x from the data given below

Price (Rs.)	10	12	13	12	16	15
Amount demanded	40	38	43	45	37	43

Estimate the likely demand when the price is Rs.20.

Solution.

x: Prices

y: Amount demanded

Regression equation of y on x is $y - \overline{y} = r\frac{\sigma_y}{\sigma_x}(x - \overline{x})$…(1)

Mean of x (Price), $\overline{x} = \frac{10+12+13+12+16+15}{6} = 13$

Mean of y (Amount demanded), $\overline{y} = \frac{40+38+43+45+37+43}{6} = 41$

Price (x)	**Amount demanded (y)**	$d_x = x - A$	$d_y = y - B$	$d_x d_y$	d_x^2	d_y^2
10	40	-3	0	0	9	0
12	38	-1	-2	2	1	4
13	43	0	3	0	0	9
12	45	-1	5	-5	1	25
16	37	3	-3	-9	9	9
15	43	2	3	6	4	9
		$\sum d_x = 0$	$\sum d_y = 6$	$\sum d_x d_y = -6$	$\sum d_x^2 = 24$	$\sum d_y^2 = 56$

Here $n = 6$

Standard deviation of x,

$$\sigma_x = \sqrt{\frac{\sum d_x^2}{n} - \left(\frac{\sum d_x}{n}\right)^2} = \sqrt{\frac{24}{6} - \left(\frac{0}{6}\right)^2} = 2$$

Standard deviation of y,

$$\sigma_y = \sqrt{\frac{\sum d_y^2}{n} - \left(\frac{\sum d_y}{n}\right)^2} = \sqrt{\frac{56}{6} - \left(\frac{6}{6}\right)^2} = 2.89$$

Coefficient of correlation,

$$r = \frac{n\sum d_x d_y - \sum d_x \sum d_y}{\sqrt{\left[\{n\sum d_x^2 - (\sum d_x)^2\}\left\{n\sum d_y^2 - \left(\sum d_y\right)^2\right\}\right]}} = \frac{6(-6) - 0(6)}{\sqrt{[6(24) - 0^2][6(56) - 6^2]}} = -0.17$$

Substituting the above values in (1), we get

$$y - 41 = \frac{(-0.17)(2.89)}{2}(x - 13) \Rightarrow y - 41 = -0.25(x - 13) \Rightarrow y = -0.25x + 44.25$$

When $x = 20, y = -0.25(20) + 44.25 = 39.25$

When the price is Rs.20, the likely demand is 39.25.

Example 4: Price indices of cotton and wool are given below for the 12 months of a year. Obtain the equations of lines of regression between the indices

Price index of cotton (x)	78	77	85	88	87	82	81	77	76	83	97	93
Price index of wool (y)	84	82	82	85	89	90	88	92	83	89	98	99

Solution.

Mean of x, $\overline{x} = \frac{78+77+85+88+87+82+81+77+76+83+97+93}{12} = \frac{1004}{12} = 83.67$

Mean of y, $\overline{y} = \frac{84+82+82+85+89+90+88+92+83+89+98+99}{12} = \frac{1061}{12} = 88.4$

Assumed mean of $x, A = 83$

Assumed mean of $y, B = 88$

Price index of cotton (x)	**Price index of wool (y)**	$d_x = x - A$	$d_y = y - B$	$d_x d_y$	d_x^2	d_y^2
78	84	-6	-4	24	36	16
77	82	-7	-6	42	49	36
85	82	1	-6	-6	1	36
88	85	4	-3	-12	16	9
87	89	3	1	3	9	1
82	90	-2	2	-4	4	4
81	88	-3	0	0	9	0
77	92	-7	4	-28	49	16
76	83	-8	-5	40	64	25
83	89	-1	1	-1	1	1
97	98	13	10	130	169	100
93	99	9	11	99	81	121
$\sum x = 1004$	$\sum y = 1061$	$\sum d_x = -4$	$\sum d_y = 5$	$\sum d_x d_y = 287$	$\sum d_x^2 = 488$	$\sum d_y^2 = 365$

Here $n = 12$

Standard deviation of x,

$$\sigma_x = \sqrt{\frac{\sum d_x^2}{n} - \left(\frac{\sum d_x}{n}\right)^2} = \sqrt{\frac{488}{12} - \left(\frac{-4}{12}\right)^2} = 6.37$$

Standard deviation of y,

$$\sigma_y = \sqrt{\frac{\sum d_y^2}{n} - \left(\frac{\sum d_y}{n}\right)^2} = \sqrt{\frac{365}{12} - \left(\frac{5}{12}\right)^2} = 5.50$$

Coefficient of correlation,

$$r = \frac{n\sum d_x d_y - \sum d_x \sum d_y}{\sqrt{\left[\{n\sum d_x^2 - (\sum d_x)^2\}\left\{n\sum d_y^2 - \left(\sum d_y\right)^2\right\}\right]}}$$
$$= \frac{12(287) - (-4)(5)}{\sqrt{[12(488) - (-4)^2][12(365) - 5^2]}} = 0.69$$

Regression equation of x on y is

$$x - \overline{x} = r\frac{\sigma_x}{\sigma_y}(y - \overline{y}) \Rightarrow x - 83.67 = \frac{(0.69)(6.37)}{5.50}(y - 88.41)$$
$$\Rightarrow x - 83.67 = 0.766(y - 88.41) \Rightarrow x = 0.799y + 13.03$$

Regression equation of y on x is

$$y - \overline{y} = r\frac{\sigma_y}{\sigma_x}(x - \overline{x}) \Rightarrow y - 88.41 = \frac{(0.69)(5.50)}{6.37}(x - 83.67)$$
$$\Rightarrow y - 88.41 = 0.599(x - 83.67) \Rightarrow y = 0.599x + 38.29$$

Angle between two regression lines :-

Note :-

1) If $r = 0$, $\tan\theta = \left(\frac{1-0}{0}\right)\left(\frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2}\right) = \infty$

$\theta = \tan^{-1}\infty = \frac{\pi}{2}$.

i.e., If $r = 0$ then $\theta = \frac{\pi}{2}$.

2) If θ is acute, then $\tan\theta = \frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2}\left(\frac{1 - r^2}{r}\right)$

) If θ is obtuse then $\tan\theta = \frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2}\left(\frac{r^2 - 1}{r}\right)$

) If $r = \pm 1$, $\theta = 0$ or π.

Problems:-

1) If $\sigma_x = \sigma_y = \sigma$ and the angle between the regression lines is $\tan^{-1}\frac{4}{3}$. Find r.

Solution: Angle between the regression lines is

$$\tan\theta = \frac{1-r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \quad \text{—①}$$

Given that

$$\theta = \tan^{-1}\frac{4}{3}, \quad \sigma_x = \sigma_y = \sigma$$

From ①, $\theta = \tan^{-1}\left(\frac{1-r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}\right)$

$$\tan^{-1}\frac{4}{3} = \tan^{-1}\left(\frac{1-r^2}{r} \cdot \frac{\sigma \cdot \sigma}{\sigma^2 + \sigma^2}\right)$$

$$\tan^{-1}\frac{4}{3} = \tan^{-1}\left(\frac{1-r^2}{r} \cdot \frac{\sigma^2}{2\sigma^2}\right)$$

$$\therefore \frac{1-r^2}{2r} = \frac{4}{3}.$$

$$3 - 3r^2 = 8r$$

$$3r^2 + 8r - 3 = 0.$$

$$3r^2 + 9r - r - 3 = 0$$

$$3r(r+3) - 1(r+3) = 0$$

$$(r+3)(3r-1) = 0$$

$$r = -3, \quad 3r - 1 = 0 \Rightarrow r = \frac{1}{3}.$$

$$\therefore r = \frac{1}{3} \quad [\because -1 \le r \le 1]$$

2) If θ is angle between two regression lines and S.D of y is twice the S.D of x and $r = 0.25$, find $\tan\theta$.

2) Solution :-

Given that $\sigma_y = 2\sigma_x$, $r = 0.25$

Angle between the regression lines is

$$\tan\theta = \frac{1-r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

$$= \frac{1-(0.25)^2}{0.25} \cdot \frac{\sigma_x \cdot 2\sigma_x}{\sigma_x^2 + 4\sigma_x^2}$$

$$= \frac{1-(0.25)^2}{0.25} \times \frac{2\sigma_x^2}{5\sigma_x^2}$$

$$= \frac{2}{5} \times \frac{1-(0.25)^2}{0.25}$$

$$= 1.5$$

$\therefore \tan\theta = 1.5$